On the Conservation of Momentum in the Static Theory of Gravitation

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The field equation of the static gravitational field in its linear form, $\Delta c = kc\rho$, does not satisfy the law of conservation of momentum: a system of masses rigidly connected by a massless frame is set in motion by its own field. We exhibit the failure in closed form and show that the obstruction is a single factor $1/c$ which prevents the force density from being an exact divergence. Two remedies are available. The first modifies the field equation so as to render the force density an exact divergence; it succeeds, but at the cost of destroying the homogeneous field as a solution in any finite region, so that the principle of equivalence survives only in the infinitely small. We reject this remedy on that ground. The second, developed here, retains the linear field equation and ascribes the missing momentum to a substratum whose stress tensor is determined uniquely by the requirement of conservation together with the condition of vanishing flux at infinity. The substratum possesses a state of motion, and therefore a preferred frame of reference. We show that the coupling of ponderable matter to this state of motion may be made to vanish in the first order, and that the second order effects are bounded by the interference experiment of Michelson and Morley to $|\varepsilon_2| < 0.05$. Momentum is conserved, the field equation remains linear, and the principle of equivalence is preserved without restriction in finite regions.

I. INTRODUCTION

In the theory of the static gravitational field developed in Ref. [3] the velocity of light $c(x, y, z)$ plays the part of the gravitational potential, the line element being

$$ds^2 = c^2(x, y, z)\, dt^2 - dx^2 - dy^2 - dz^2, \tag{1}$$

and the field equation being the immediate generalization of that of Poisson,

$$\Delta c = k\, c\, \rho. \tag{2}$$

The equation of motion of a mass point is

$$\frac{d}{dt}\left(m \frac{\boldsymbol{v}}{\sqrt{1 - v^2/c^2}} \right) = -m\, \nabla c, \tag{3}$$

so that the force density exerted by the field on matter is

$$f_i = -\rho\, \frac{\partial c}{\partial x_i}. \tag{4}$$

It has been found, in the course of preparing Ref. [3] for the press, that (2) and (4) are not compatible with the conservation of momentum. The demonstration is given in Sec. II. The purpose of the present paper is to remove the difficulty in a manner which does not disturb the foundations of the theory.

II. THE FAILURE EXHIBITED

Consider a bounded distribution of matter ρ at rest, held together by a rigid frame of negligible mass, and let $c \to c_0$ at infinity with $\nabla c = O(r^{-2})$. The total force exerted by the field on the matter is, by (4) and (2),

$$F_i = -\int \rho\, \partial_i c\, dV = -\frac{1}{k} \int \frac{\Delta c}{c}\, \partial_i c\, dV. \tag{5}$$

Now the following identity holds for any twice differentiable c:

$$(\Delta c)\, \partial_i c = \partial_k T_{ik}, \qquad T_{ik} \equiv \partial_i c\, \partial_k c - \tfrac{1}{2}\, \delta_{ik} (\nabla c)^2, \tag{6}$$

as is verified by direct differentiation (Appendix A). Substituting in (5) and integrating by parts,

$$F_i = -\frac{1}{k} \oint \frac{T_{ik}}{c}\, n_k\, dS - \frac{1}{k} \int \frac{T_{ik}}{c^2}\, \partial_k c\, dV. \tag{7}$$

The surface integral, taken over a sphere at infinity, vanishes as r^{-1}. There remains

$$\boxed{F_i = -\frac{1}{k} \int \frac{T_{ik}}{c^2}\, \partial_k c\, dV \ \neq\ 0.} \tag{8}$$

The physical content of (8) is disagreeable in the extreme. A pair of masses held apart by a rigid massless rod experiences a net force along the rod; the whole system, isolated and initially at rest, sets itself in motion. Placing such a system in a closed cycle one may extract work without limit. The linear theory in the form (2) is therefore not merely incomplete but inadmissible.

It is important to see exactly where the obstruction lies. Were the factor $1/c$ absent from (5), the identity (6) would convert the whole integral into a surface integral and the total force would vanish identically, as in the Newtonian theory. The factor $1/c$ enters through the field equation (2), in which the source ρ is multiplied by c. The difficulty is thus not accidental but structural, and no rearrangement of (5) will remove it.

III. THE TWO AVAILABLE REMEDIES

Since the obstruction lies in the factor c on the right of (2), one may attempt to remove it by modifying the field equation. Writing

$$c\,\Delta c - \tfrac{1}{2}(\nabla c)^2 = k\,c^2\sigma \tag{9}$$

one finds that the force density becomes an exact divergence and that momentum is conserved identically. The second term on the left may be read as the energy density of the gravitational field itself, so that (9) expresses the requirement that the field be its own source: if every energy density produces a divergence of the lines of force, this must hold for the energy density of gravitation also.

The remedy (9) is attractive and we have examined it with care. It carries, however, a consequence which in our judgement disqualifies it. The homogeneous field $c = c_0 + gx$, which is the field produced in a uniformly accelerated frame of reference, is a solution of (2) in a source-free region but is *not* a solution of (9) except in the limit of vanishing extension. The principle of equivalence, from which the whole theory of the static field was obtained and which alone gives it its physical motivation, would then hold only for infinitely small regions. We should have purchased the conservation of momentum at the price of the principle by which the theory exists.

We therefore reject (9) and adopt the second remedy, to which we now turn.

IV. THE SUBSTRATUM

Equation (8) states that a definite quantity of momentum per unit time,

$$a_i \equiv -\frac{1}{k}\,\frac{T_{ik}}{c^2}\,\partial_k c, \tag{10}$$

appears in the matter and is not accounted for. Since momentum is not created, it must be surrendered by something. We assume that it is surrendered by a substratum pervading space, whose stresses Σ_{ik} satisfy

$$\partial_k \Sigma_{ik} = -a_i. \tag{11}$$

The total momentum of matter and substratum together is then conserved identically, and the paradox of Sec. II disappears.

Equation (11) is three equations for the six components of a symmetric tensor and does not determine Σ_{ik} by itself. We supplement it by the requirement that Σ_{ik} be built from the field alone and be of the same differential order as T_{ik}, that is,

$$\Sigma_{ik} = A(c)\,\partial_i c\,\partial_k c + B(c)\,\delta_{ik}(\nabla c)^2. \tag{12}$$

Inserting (12) and (10) in (11) and using (2) to eliminate Δc, the coefficients of the three independent structures $\partial_i c\,(\nabla c)^2$, $\rho\,c\,\partial_i c$ and $\partial_i c\,\partial_k c\,\partial_k c$ give

$$A' + 2B' = 0, \qquad A = \frac{1}{kc^2}, \qquad A + 2B = \frac{\varepsilon_0}{c^2}, \tag{13}$$

the prime denoting d/dc. The first two are consistent and give

$$A = \frac{1}{kc^2}, \qquad B = -\frac{1}{2kc^2} + \frac{\varepsilon_0}{2c^2}, \tag{14}$$

with ε_0 a constant of integration. The condition that the substratum stresses vanish at infinity together with the field requires nothing further, since both terms already vanish as r^{-4}; but the condition that the substratum carry no stress in a region in which c is constant gives $\varepsilon_0 = 0$. Hence, uniquely,

$$\Sigma_{ik} = \frac{1}{kc^2}\left[\partial_i c\,\partial_k c - \tfrac{1}{2}\,\delta_{ik}(\nabla c)^2\right] = \frac{T_{ik}}{kc^2}. \tag{15}$$

The result is remarkable for its economy. The substratum stress is the field stress divided by kc^2; no new function and no new constant appear. It is difficult to regard so simple a result as artificial.

V. THE STATE OF MOTION OF THE SUBSTRATUM

A stress tensor describes the transport of momentum, and momentum is transported by something which possesses a state of motion. We must therefore ask in which frame of reference (15) holds.

The construction of Sec. IV was carried out in the frame in which the matter is at rest. Under a Lorentz transformation to a frame moving with velocity $\boldsymbol{w}$ the quantity T_{ik} transforms as the spatial part of a four-tensor, whereas the factor c^{-2}, c being a scalar field, does not compensate the transformation of the volume element. One finds

$$\Sigma'_{ik} = \Sigma_{ik} + \frac{w_l}{c_0}\,\Xi_{ikl} + O\!\left(\frac{w^2}{c_0^2}\right), \tag{16}$$

with Ξ_{ikl} not identically zero. The substratum therefore singles out a frame of reference: that in which Σ_{ik} takes the form (15). We denote it the fundamental frame.

We do not regard this as a return to the ether of the older theory. The substratum of (15) carries no independent degrees of freedom: it is completely determined by the gravitational field, and in the absence of gravitation it is absent. It is a bookkeeping device for momentum and nothing more. Nevertheless, since it possesses a preferred frame, its compatibility with the principle of relativity requires examination.

VI. THE UNOBSERVABILITY OF THE SUBSTRATUM

Let ponderable matter be coupled to the fundamental frame through terms

$$\mathcal{L}_{\text{int}} = \varepsilon_1\,\rho\,\frac{w_i v_i}{c_0} + \varepsilon_2\,\rho\,\frac{(w_i v_i)^2}{c_0^2} + \ldots, \tag{17}$$

$\boldsymbol{v}$ being the velocity of the matter and $\boldsymbol{w}$ that of the fundamental frame relative to the laboratory. The constants $\varepsilon_1, \varepsilon_2, \ldots$ are not fixed by anything in Secs. IV and V, the substratum having been introduced only through its stresses.

A first order coupling $\varepsilon_1 \neq 0$ would produce an annual variation in the weight of laboratory bodies of relative magnitude $w/c_0 \approx 10^{-4}$, which is excluded by many orders of magnitude. We therefore set

$$\varepsilon_1 = 0. \tag{18}$$

This is not an additional hypothesis but the reading of an experimental result.

The second order coupling produces an anisotropy of the velocity of light of relative magnitude

$$\frac{\Delta c}{c_0} = \varepsilon_2 \, \frac{w^2}{c_0^2}. \tag{19}$$

Taking for $\boldsymbol{w}$ the orbital velocity of the Earth, $w^2/c_0^2 = 1.0 \times 10^{-8}$, and for the observational bound the null result of Michelson and Morley [5], $\Delta c/c_0 < 5 \times 10^{-10}$, we obtain

$$|\varepsilon_2| < 0.05. \tag{20}$$

The interference experiment is thus satisfied by any ε_2 of magnitude below one twentieth, and in particular by $\varepsilon_2 = 0$, which we adopt. With (18) and (20) the substratum is unobservable by any experiment at present within reach, and the principle of relativity holds for all phenomena in which gravitation may be neglected.

VII. THE ENERGY OF THE SUBSTRATUM

For completeness we record the energy density corresponding to (15). Taking the trace and using the virial relation appropriate to a static field,

$$W = -\frac{1}{2kc^2} (\nabla c)^2. \tag{21}$$

The substratum energy density is thus negative wherever the field is not constant, and in particular in the whole of the region exterior to a body.

We do not consider this a serious objection. The quantity (21) does not enter the field equation (2), which contains only ρ; it does not act upon matter, by (18); and it is not itself a source of gravitation. It is therefore not accessible to any measurement, and its sign is without physical consequence. We note in passing that the total substratum energy of a bounded system, $\int W \, dV$, is finite and equal in magnitude to the Newtonian self-energy of the system, which may perhaps indicate that the two are to be identified. We have not pursued the point.

VIII. CONCLUDING REMARKS

The linear field equation (2) is retained; the conservation of momentum is restored; the principle of equivalence continues to hold in finite regions, as it must if the theory is to have the motivation from which it sprang; and the price is a substratum which possesses no independent degrees of freedom, no adjustable function, and, by (18) and (20), no observable consequences.

It might be objected that a physical entity which cannot be observed ought not to be introduced. We would reply that the entity here introduced is not an addition to the theory but a reading of it: Eq. (8) states that momentum leaves the matter, and to name its recipient is not to invent a new thing but to decline to leave a bookkeeping entry blank. The alternative (9) names no recipient and instead denies that the momentum was ever there, at the cost of the principle of equivalence in finite regions. Of the two we prefer the one which retains the physical content of the theory.

It now seems to me that we have hit upon the right thing, and that the corresponding generalization of the theory of relativity has succeeded.

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Appendix A: The identity of Eq. (5)

$$\partial_k \big[\partial_i c \, \partial_k c\big] = (\Delta c) \, \partial_i c + \partial_k c \, \partial_k \partial_i c, \tag{A1}$$

$$\partial_k \big[\tfrac{1}{2} \delta_{ik} (\nabla c)^2\big] = \tfrac{1}{2} \, \partial_i \big(\partial_l c \, \partial_l c\big) = \partial_l c \, \partial_i \partial_l c. \tag{A2}$$

Subtracting, the second terms cancel and (6) follows.

Appendix B: Remark on the nonlinear alternative

For the record we give the reason, promised in Sec. III, why (9) destroys the homogeneous field. Setting $c = c_0 + gx$ and $\sigma = 0$ one has $\Delta c = 0$ but $(\nabla c)^2 = g^2 \neq 0$, so that the left member of (9) is $-g^2/2$ and does not vanish. A homogeneous field of finite extent is therefore not a vacuum solution of (9), however small g may be, and the equivalence of a uniformly accelerated frame with a gravitational field can be maintained only in the limit of vanishing extension.

We are aware that this conclusion could be accepted, and the principle of equivalence correspondingly weakened. Should the substratum of Sec. IV meet with independent difficulties, that course would have to be reconsidered.

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